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ORIGINAL ARTICLE

Application of wearable perception and soft computing algorithms in personalized recommendation of music online teaching

Dai Yufei[1]

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Abstract
With the widespread application of wearable technology, the field of music education has ushered in new opportunities for personalized recommendations. However, the traditional music teaching model has shortcomings such as low level of informatization and single teaching content. This study aims to utilize wearable sensor technology and soft computing algorithms to explore how to achieve personalized recommendations for music information technology teaching, thereby improving teaching effectiveness and user experience. Research on selecting suitable wearable sensors to obtain physiological indicators and exercise data of students during music learning, monitor student information in real-time, and utilize soft computing algorithms for processing. By analyzing and mining data, reveal patterns such as emotional changes and fluctuations in focus among students during the learning process. Finally, by integrating sensor data and the results of soft computing algorithms, a corresponding algorithm model is designed, taking into account the personality characteristics, learning preferences, and characteristics of music content of students, in order to achieve personalized recommendations. By continuously optimizing model parameters and algorithm logic, the accuracy and personalization of recommendations are improved. The experiment verified the effectiveness of personalized recommendation based on wearable perception and soft computing algorithms in music teaching, which can better meet the personalized needs of students, improve teaching effectiveness and learning motivation.

Keywords Wearable perception · Soft computing algorithms · Music information technology teaching · Personalized recommendations

1 Introduction

Personalized recommendations can improve teaching level, promote personal development of students, and cultivate innovative talents (Lee and Brusilovsky 2017). However, traditional teaching methods mainly focus on classroom teaching and training, which to some extent affects the efficiency and interest of students in learning. With the development of educational technology, the teaching philosophy and methods of sight singing and ear training are also constantly innovating (Lin et al. 2017). Despite the use of multimedia tools and teaching software, it still cannot meet the needs of intelligent teaching. Although existing online courses and learning platforms provide abundant learning resources and different learning approaches, they still lack personalized and intelligent teaching support.

In this article, a music information teaching and personalized recommendation system based on wearable sensing and soft computing algorithms is designed, providing feasible solutions for music information teaching, personalized teaching resource push, and ability evaluation (Yang 2022). This system utilizes the perception function of wearable devices to monitor the status and behavior of students in real-time during music learning, and analyzes and processes data through soft computing algorithms. By collecting student behavior data and learning characteristics, the system can gain a deeper understanding of each student's learning habits, preferences, and levels. Based on this information, the system can provide personalized learning resource recommendations, recommending suitable learning materials and music for students according to their interests and abilities. In this way, students can better immerse themselves

✉ Dai Yufei
Dlpiano123@163.com

[1] Xinzhou Normal University, Xinzhou 034000, Shanxi, China

in the field of music that they are interested in, and gain targeted learning experiences, improving learning effectiveness and interest. The system also has the function of capability evaluation (Breysse et al. 2022). By analyzing the data of students during their learning process, the system can evaluate their musical abilities and progress, providing timely feedback and guidance to teachers. Teachers can adjust their teaching plans and methods based on the evaluation results of students, better meet their needs, and provide personalized guidance and support. The design and application of this music information technology teaching and personalized recommendation system can not only provide intelligent and personalized teaching support for music education, but also stimulate students' interest and initiative in learning (Lu and Deng 2021). It also provides effective tools and resources for teachers to better guide their learning and promote their development.

2 Related work

In terms of domain knowledge models, the literature has established a knowledge system covering music theory, sight singing training methods, and score interpretation (Chai et al. 2018). This model should include basic music knowledge and skill points, as well as score features and rules related to sight singing training. By structuring and categorizing these knowledge, the system can better understand the disciplinary characteristics of music sight singing, providing learners with systematic teaching content and guidance. In terms of learner models, the literature considers individual differences and learning needs of students to construct a personalized learner model (Zhou et al. 2021). This model should comprehensively consider information such as students' music background, learning goals, learning styles, and cognitive characteristics, to help the system better understand the needs and characteristics of each learner. By analyzing and mining student learning data, the system can tailor learning paths and content for each student, provide personalized learning support and feedback, and promote their progress and improvement in music sight singing. In terms of teacher models, literature has designed an effective teacher support system to help teachers better manage and guide students in their learning process (Reitman and Karge 2019). This teacher model should include functional modules such as teaching resource management, student performance evaluation, and teaching plan design, supporting teachers to monitor and analyze student learning situations, and providing real-time teaching feedback and guidance. By combining learner models and domain knowledge models, teachers can better personalize teaching content and methods, promoting the comprehensive development of students in the field of music sight singing. The literature proposes a comprehensive and specific set of score difficulty evaluation indicators by analyzing the characteristics of different scores in pitch, rhythm complexity, beat variation, music structure, and other aspects (Hirosawa et al. 2022). These indicators can better reflect the difficulty level of score, laying the foundation for subsequent score recommendation algorithms. Based on these difficulty characteristics of music scores, the literature has designed an intelligent music score recommendation algorithm and strategy (Tang and Zhang 2022). This algorithm combines the personalized needs of learners and the difficulty characteristics of sheet music, and utilizes machine learning and data mining techniques to analyze and model the difficulty preferences of learners, thereby achieving targeted sheet music recommendations. Through intelligent computing and recommendation systems, learners can obtain score resources that better meet their needs and levels during the learning process, improving learning efficiency and satisfaction (Liang et al. 2018).

Through in-depth analysis of the core elements of music sight singing ability, such as pitch accuracy, rhythm, and melody memory, a complete evaluation system for music sight singing ability has been established in the literature (Zhe 2021). In this system, multiple ability evaluation indicators are covered, which can comprehensively and accurately reflect the learner's level of sight singing. Based on these evaluation indicators of music sight singing ability, an intelligent evaluation model was designed in the literature (Cheng et al. 2020). This model combines music professional knowledge and computer technology, using data analysis and pattern recognition algorithms to accurately evaluate the sight singing ability of learners. By collecting data on the performance of learners in sight singing and matching it with the model for analysis, the system can accurately evaluate the performance level of learners on different ability indicators, help teachers better understand the learning status of students, and provide strong support for personalized teaching and guidance. The literature has achieved intelligent evaluation of learners' cognitive level in music information technology teaching and personalized recommendation systems (Xu 2019). Teachers and systems can provide personalized learning resources and guidance plans based on the evaluation results of students, helping them better improve their sight singing ability and achieve personalized learning goals. This intelligent evaluation system has introduced new ideas and methods into the field of music education, promoting the continuous improvement of students' learning effectiveness and experience (Greer and Mark 2016). The literature analyzes a large number of musical works and regards tonality as an important indicator of complexity. The judgment of tonality is mainly achieved by analyzing the correction of prosodic position and pitch duration in music. Research has found that the complexity of tonality is closely related to the number and intensity of corrections, and can be used as an important indicator for predicting music complexity. The literature focuses on the interval principle of range direction and range differences in music. The direction of range refers to the movement direction of pitch in music, while the principle of interval in range differences refers to the frequency and amplitude of range changes. Research has found that these factors play an important role in predicting music complexity. The more frequent the changes in range direction, the greater the intermittency of range differences, and music is considered to have higher complexity. The literature studied the impact of rhythm principle segmentation and rhythm variability on music complexity judgment. The principle of rhythm segmentation refers to the structure and segmentation of rhythm in music, while rhythm variability reflects the degree of rhythm variation and irregularity (Pandeya et al. 2022). Research has found that the segmentation of rhythm principles and the increase in rhythm variability are related to the complexity of music and can be used as important indicators for predicting music complexity. Learning level characteristics refer to the level of mastery of learners in a specific field or discipline (Visser et al. 2018). In order to obtain the learning level characteristics of learners, literature uses data collected from learners' exam scores, online test results, or other evaluation tools. These data help the system understand the strengths and weaknesses of learners, thereby recommending teaching resources of appropriate difficulty to them. The literature suggests that learning style characteristics refer to the preferred ways or strategies that learners use during the learning process (Samarakou et al. 2018). Learners may have different learning styles, such as visual, auditory, and hands-on. The SAELS system can obtain the learning style characteristics of learners through questionnaires, observing their behavior during the learning process, or other methods. These features help the system understand learners' preferences for different types of teaching resources, thereby providing them with resource recommendations that are more in line with their learning style. After integrating the learning level and learning style characteristics of learners, the SAELS system in the literature utilizes machine learning or data mining techniques to train models to predict learners' interest or fitness for different resources. Then, the system can recommend the most relevant and suitable resources to learners based on their personalized needs and interests, in order to improve learning effectiveness and learner engagement.

3 Wearable sensing and data collection

3.1 Embedded wearable system architecture

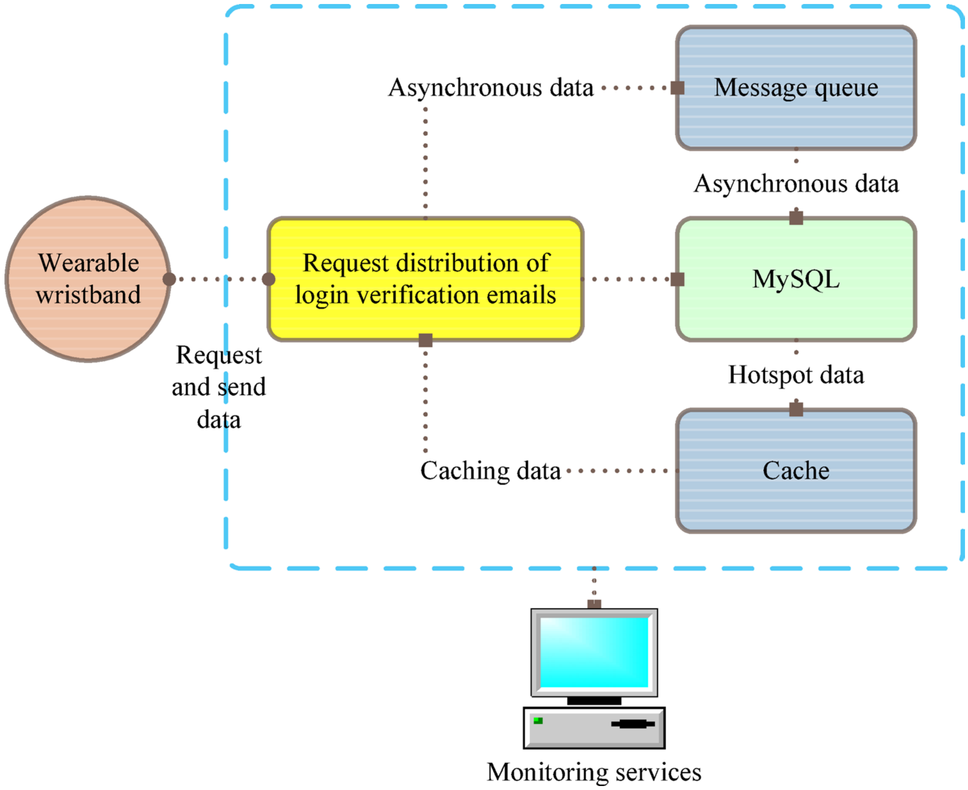
The software functional diagram in Fig. 1 shows the overall architecture of the embedded system and the relationships between modules. In order to optimize the utilization efficiency of CPU computing resources, this design adopts the open-source embedded real-time operating system RT Thread. RT Thread has the characteristics of lightweight, fast response, and scalability, making it suitable for application in embedded systems. By using RT Thread, this design breaks down multiple functional modules of the system into multiple threads to run, each responsible for different tasks such as data collection, sensor control, user interface processing, etc., running independently of each other, and communicating and synchronizing through message queues or event mechanisms.

In this system, there are multiple threads responsible for different functions. The audio thread uses MEMS dual microphones for voice acquisition and encodes and converts the simulated audio signal. Use the audio decoding chip WM8978 for signal filtering, and set the frequency of voice data acquisition to 8000 Hz. Compute threads to extract key features from audio signals. This thread processes voice data every 10 min and uses a dual buffering strategy to ensure synchronization between voice and data collection. After processing, the original voice data will be automatically deleted. The extracted speech features include entropy spectrum, energy, brightness, and resonance peaks. The collected data will be formatted for convenient subsequent storage and transmission. The sampling rate of this thread is 100 Hz. The environmental sensor thread collects environmental characteristics, including data such as light, temperature, and air pressure. The collected data also needs to be formatted. The sampling rate of this thread is 1 Hz. The file thread notifies the file thread to save a certain amount of data collected by the action sensor and environment sensor through a signaling mechanism. OLED threads display real-time collected data, battery level, and time information on a 128 * 64 OLED LCD screen. The human–computer interaction thread is equipped with three buttons, and the wearer can interact with the control device through the human–computer interaction thread. These threads work together to achieve the system's functions of collecting, processing, saving, and displaying audio and sensor data.

3.2 Voice data processing

In order to protect user privacy, two MEMS microphones were used to collect voice analog signals for this audio data collection. In terminal devices, in order to maximize

Fig. 1 Function diagram of wearable wristband software



the protection of user privacy, the original voice signals are processed. Only a small amount of feature information was extracted for preservation, rather than storing the complete speech signal. These features include entropy spectrum, energy, brightness, and resonance peaks. Through this approach, the demand for storage space can be reduced and sensitive information of users can be appropriately protected. The pre emphasis transfer function can be expressed using formula (1):

$$\mathrm{H(z) = 1 - aZ^{-1}} \tag{1}$$

where z is the complex variable in the frequency domain, α is the preaccentuated coefficient of the preaccentuated filter.

By pre emphasis, a rising slope can be introduced into the spectrum of the speech signal, which means that the gain in the high-frequency part is higher than that in the low-frequency part. This can reduce losses caused by high-frequency signal attenuation during transmission and recording, and increase the energy of the high-frequency part, thereby improving the quality and resolution of the signal. In actual code, pre emphasis is usually achieved through a difference equation, which can be expressed as formula (2):

$$\mathrm{y(t) = x(t) - ax(t-1)} \tag{2}$$

Among them, y[n] is the pre emphasis signal:

$$\mathrm{y(n) = \sum_{n=-\infty}^{\infty} x(m)w(n-m)} \tag{3}$$

The windowing operation is also a commonly used processing method in spectrum analysis, and there are many common window functions to choose from, among which the Hanning window is suitable for the vast majority of situations. The calculation formula for Hanning window is:

$$\mathrm{w\,(n)} = \begin{cases} 0.54 - 0.46{*}\cos\left(\frac{2\pi n}{L-1}\right) & 0 \le n \le N-1 \\ 0 & \text{other} \end{cases} \tag{4}$$

Among them, w [n] is the value of the Hanning window, and N is the window length. The calculation of Hanning windows can help with windowing the signal. The calculation process of short-term energy is as follows:

Firstly, preprocess each frame of speech data to obtain the processed speech signal y_i.

Then, for each y_i, calculate its energy E(i). The calculation of energy can be obtained by taking the square of each sample value of the speech signal and summing it up. Assuming that each frame of speech signal has N sample points, the formula for calculating energy E(i) is as follows:

$$\mathrm{E(i) = \sum_{n=0}^{N-1} y_i^2(n), i = 1,2,\cdots,M} \tag{5}$$

By performing Fourier transform on each y_i, the spectrum is obtained. Then, based on the spectrum, calculate the probability density P(i) of the spectrum. The calculation formula for spectral probability density is as follows:

$$p_i(n) = \frac{Y_i(n)}{\sum_{n=0}^{N/2} Y_i(n)} \tag{6}$$

Calculate the short-term spectral entropy EP_i based on the spectral probability density P(i). The calculation formula for short-term spectral entropy is as follows:

$$EP_i = -\sum_{n=0}^{N/2} p_i(n) \log p_i(n) \tag{7}$$

Finally, the formula for calculating discrete brightness is as follows:

$$DB_i = \frac{\sum_{n=0}^{N/2} w_n \cdot |y_n|^2}{\sum_{n=0}^{N/2} |y_n|^2} \tag{8}$$

3.3 Sensing data recognition

Time domain features describe the statistical characteristics of acceleration data over a period of time, which can be used to analyze the statistical characteristics and volatility of acceleration data.

For the mean and root mean square (RMS) of acceleration data, their calculation formulas are as follows:

$$\overline{X} = \frac{1}{T} \sum_{t=1}^{T} x_t \tag{9}$$

$$RMS(X) = \sqrt{\frac{1}{T} \sum_{t=1}^{T} x_t^2} \tag{10}$$

Both of these features reflect the overall numerical size of the acceleration data, but the root mean square values the volatility of the data more than the simple mean. Variance is used to measure the average square of the difference between the data point and its mean. The calculation formula is shown in (11):

$$\delta_X = \frac{1}{T-1} \sum_{t=1}^{T} \left(x_t - \overline{X}\right)^2 \tag{11}$$

Absolute Mean Difference (MAD) measures the degree of fluctuation in acceleration data and is the average absolute difference between data points and the mean. The calculation formula is as follows (12):

$$MAD(X) = \frac{1}{T} \sum_{t=1}^{T} \left|x_t - \overline{X}\right| \tag{12}$$

The correlation coefficient between the random variables X and Y with different axis accelerations is:

$$\text{correlation}(X, Y) = \frac{\text{cov}(X, Y)}{\delta_X \delta_Y} \tag{13}$$

The calculation method of energy mainly involves performing discrete fast Fourier transform (DFT) on the signal, and then calculating the sum of squared amplitudes of each frequency component. The calculation formula is as follows (14):

$$\text{Energy}(X) = \frac{\sum_{i=1}^{T} |F_i|^2}{T} \tag{14}$$

By calculating correlation coefficients and energy, we can better understand the characteristics of acceleration data in the spatial domain, including the correlation between different axes and the spectral energy distribution of signals. These features can provide useful information for further behavior analysis and pattern recognition.

3.4 Heart rate sensing acquisition

In PPG signals, there is a DC component, which leads to a larger amplitude of the signal, while the range of variation of the AC component is smaller. By removing the DC signal, changes in the AC signal can be highlighted. After removing the DC signal, heart rate related AC signals can be observed more clearly. Use a low-pass filter to filter the PPG signal after removing the DC signal. The filter used this time selected an 8-pass filter, retaining low-frequency signals related to heart rate. The calculation formula for heart rate is shown in formula (15):

$$H = N \times f \times 60 \tag{15}$$

Among them, N represents the number of peaks counted in one processing, f represents the calculation frequency of heart rate, and H represents the final heart rate value. The selection of filter parameters for this time was obtained through processing on an offline PC, and then the heart rate algorithm was ported to an embedded device. On embedded

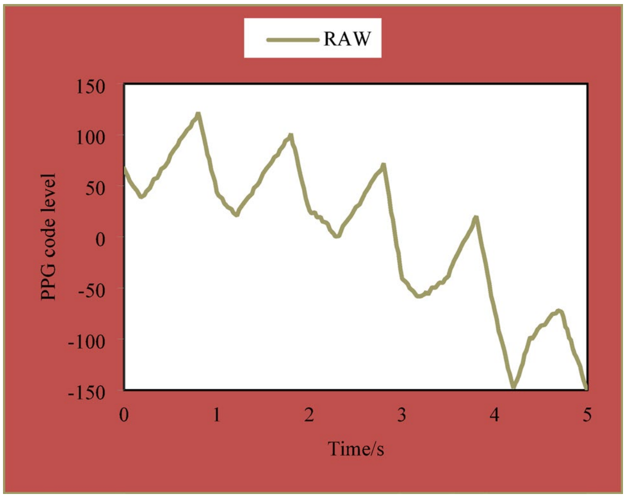


Fig. 2 Before PPG signal processing

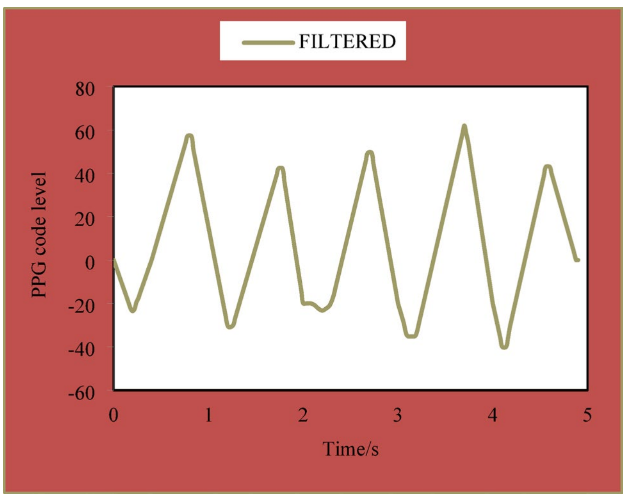


Fig. 3 After PPG signal processing

Table 1 Wearable device data collection testing

Wearing duration	Collection rates	Error rate
10 min	89.01%	0.00411%
1 h	89.08%	0.00409%
6 h	89.36%	0.00399%
12 h	88.91%	0.00400%

devices, the processing frequency of heart rate is once every 4 s (0.25 Hz), and the sampling frequency is 25 Hz. In a stationary state, the image of the original PPG signal data is shown in Fig. 2.

The data in Fig. 3 shows that some noise and interference signals have been removed, highlighting the waveform characteristics related to heart rate.

Table 2 Data transmission under stable indoor and outdoor conditions

Wearing duration	Indoor reception rate	Outdoor reception rate
10 min	98.24%	91.50%
1 h	98.84%	91.08%
6 h	96.96%	92.28%

3.5 Analysis of data collection effectiveness

During the testing process, the testers wore wearable sensors for data collection for a duration of 10 min, 1 h, 6 h, and 12 h, as shown in Table 1.

According to the data in Table 1, it can be observed that the actual frequency of data collection is slightly lower than the set frequency. This is because during data collection, additional operations such as data formatting, storage, etc. are required, which can take up some time and result in the actual data collection frequency being slightly lower than the set frequency. During the data collection process, there may also be some erroneous data, such as garbled data or other abnormal data. However, from Table 1, it can be seen that the proportion of these erroneous data is very low. Therefore, in the subsequent data processing process, the impact of these erroneous data can be ignored.

According to the data in Table 2, data transmission was observed in different environments (indoor and outdoor).

In Table 2, the reason for the decrease in outdoor data transmission reception rate is the possible base station switching problem that NB IoT modules may encounter when the outdoor activity range is large. Base station switching refers to when a user enters the coverage range of another base station from the coverage range of one base station, the device needs to establish a connection with the new base station. In a stable indoor state, the signal received by the device is strong and stable, so the data transmission reception rate is high. But when the device moves from indoors to outdoors, it may exceed the coverage range of the currently connected base station and require base station switching. During the switching process, there will be a short period of no signal state (usually at the millisecond level), which can cause interruptions in data transmission and a decrease in reception rate. To address this issue, wearable devices store data on SD cards for backup during data transmission. In this way, even if there is an interruption or failure in data transmission, the integrity and availability of the data can still be ensured by reading the backup data from the SD card. Through this approach, users can still obtain complete data records even in the event of base station switching or other communication issues.

4 Personalized recommendation strategies based on soft computing

4.1 Fundamentals of soft computing algorithms

The hard c-means (HCM) algorithm is extended to fuzzy form by introducing a membership function to weight the distance between each sample and each class prototype, thereby extending the objective function $J_1(W, P)$ of the sum of squared intra class errors to $J_2(W, P)$:

$$\begin{gathered} minJ_2(W,P) = \sum_{i=1}^{c}\sum_{j=1}^{n}(w_{ij})^2 \cdot d^2(x_j, p_i) \\ s.t. W \in M_f \end{gathered} \tag{16}$$

Formula (17) is a more general form used to describe the objective function of fuzzy clustering.

$$\begin{aligned} & \min J_\alpha(W,P) = \sum_{i=1}^{c}\sum_{j=1}^{n}(w_{ij})^\alpha \cdot d^2(x_j, p_i) \\ & \text{s.t. } W \in M_f \end{aligned} \tag{17}$$

Use formula (18) to calculate or update the classification matrix, and formula (19) is used to calculate the new cluster center, i.e. the cluster prototype matrix.

$$\mu_{ik}^{(b)} = \left\{ \sum_{j=1}^{c} \left[\frac{d_{ik}^{(b)}}{d_{jk}^{(b)}} \right]^{\frac{2}{\alpha-1}} \right\}^{-1} \tag{18}$$

$$\mu_{ir}^{(b)} = 1;\ j \neq r, \mu_{ij}^{(b)} = 0 \tag{19}$$

Formula (20) is a regularized version of Formula (19), which introduces a normalization factor when calculating cluster centers to ensure that the cluster centers meet the normalization conditions.

$$p_i^{(b+1)} = \frac{\sum_{k=1}^{n}\left(\mu_{ik}^{(b)}\right)^\alpha \cdot x_k}{\sum_{k=1}^{n}\left(\mu_{ik}^{(b)}\right)^\alpha}, i = 1, 2, \cdots, c \tag{20}$$

4.2 Collaborative filtering recommendation algorithm

The cosine similarity formula is as follows:

$$\mathrm{sim}(i,j) = \frac{\sum_{v\in U} R_{v,i}R_{v,j}}{\sqrt{\sum_{v\in U} R_{v,i}^2}\sqrt{\sum_{v\in U} R_{v,j}^2}} \tag{21}$$

The modified cosine similarity takes into account the user's rating bias, which can more accurately measure the degree of similarity between items. The formula is as follows:

$$\mathrm{sim}(i,j) = \frac{\sum_{v\in U}\left(R_{v,i} - \overline{R}_v\right)\left(R_{v,j} - \overline{R}_v\right)}{\sqrt{\sum_{v\in U}\left(R_{v,i} - \overline{R}_v\right)^2}\sqrt{\sum_{v\in U}\left(R_{v,j} - \overline{R}_v\right)^2}} \tag{22}$$

The Pearson similarity formula is as follows:

$$p(i,j) = \frac{\sum_{v\in U}\left(R_{v,i} - \overline{R}_i\right)\left(R_{v,j} - \overline{R}_j\right)}{\sqrt{\sum_{v\in U}\left(R_{v,i} - \overline{R}_i\right)^2}\sqrt{\sum_{v\in U}\left(R_{v,j} - \overline{R}_j\right)^2}} \tag{23}$$

The prediction scoring formula is as follows:

$$R_{u,j} = \overline{R}_u + \frac{\sum_{v\in U}\mathrm{sim}(i,j)\left(R_{v,j} - \overline{R}_v\right)}{\sum_{v\in U}|\mathrm{sim}(i,j)|} \tag{24}$$

4.3 Algorithm optimization

Formula (25) defines an exponential forgetting function, which is used to indicate that the weight of user interests decays over time.

$$w_T(i) = e^{-\alpha\left(i^{prime} - t_i\right)} \tag{25}$$

When calculating similarity, time factors can be considered and a forgetting function can be used to attenuate the original score. Formula (26) represents the similarity calculation method considering the forgetting function:

$$R_{u,i} = w_T(i) \times r_{u,i} \tag{26}$$

By applying attenuation weights to the rating vector, the weight of past ratings can be reduced, and more emphasis can be placed on recent ratings, thereby improving the accuracy of similarity calculation.

4.4 Analysis of personalized recommendation effectiveness

As shown in Fig. 4, the improved CBR algorithm shows a certain improvement in user recommendation accuracy compared to the original algorithm under the same conditions. This result can be explained as users having a higher interest in newer resources. Over time, user interests and preferences may change. The improved CBR algorithm attenuates ratings by introducing a forgetting function, focusing more on recent ratings and more accurately reflecting the user's current interests. In contrast, traditional CBR

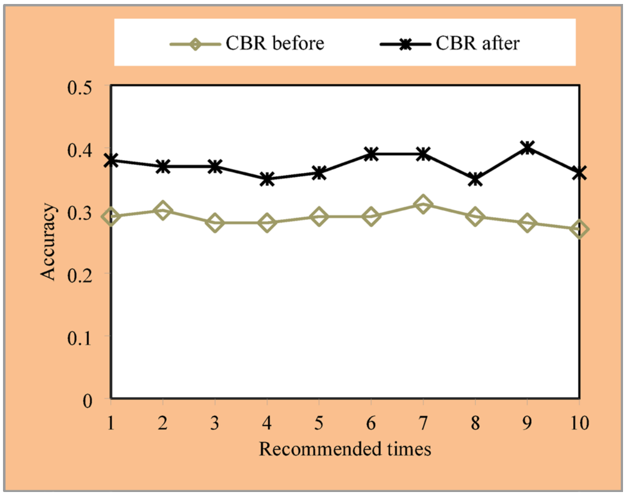


Fig. 4 Comparison of accuracy before and after improvement

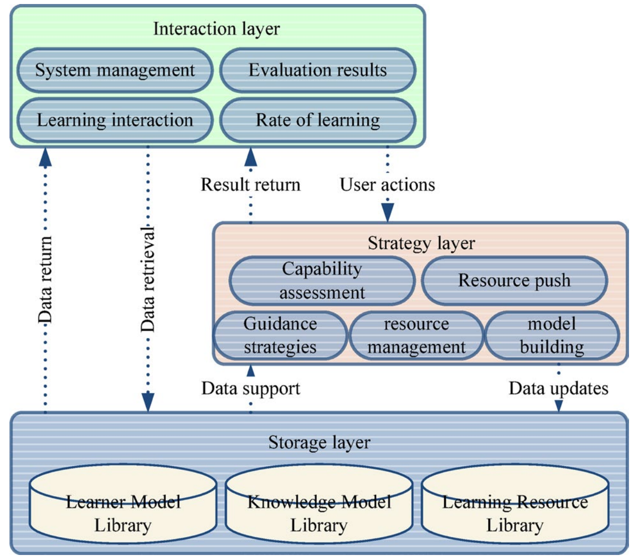


Fig. 5 Structure diagram of music information teaching and personalized recommendation system

algorithms may allocate weights to historical ratings relatively evenly, making it difficult to distinguish user current preferences. Therefore, the improved algorithm can better capture users' interest in newer resources and improve the accuracy of recommendations. This also confirms the importance of time factor in recommendation systems from another perspective, as well as the effectiveness of forgetting functions in modeling user interest decay.

5 Design of music informatization teaching and personalized recommendation system

5.1 System requirements

Personalized teaching is a teaching method adopted based on the individual characteristics of learners. It is an important educational concept and teaching mode in comprehensive quality education, and also a necessary path to achieve quality education. Personalized teaching requires educators to respect and pay attention to individual differences among students, provide personalized guidance and assistance for different cognitive levels and learning abilities. Personalized teaching requires frequent communication and interaction between teachers and students, in order to understand the learning situation, learning style, and learning confusion of learners, and provide timely guidance. In virtual learning environments, personalized teaching is mainly achieved through personalized services and support. This includes the promotion of learning resources, planning of learning paths, and diverse learning tools. Through these personalized means, students can learn according to their own interests and learning pace, achieving better learning outcomes.

In personalized recommendation systems, the role of the learner model is to analyze the learner's learning behavior, evaluate their knowledge level and skill status, and recommend content that meets their interests and learning needs. The learner model can predict the learning preferences and potential needs of learners based on their historical behavioral data, such as learning records, homework performance, etc. The association rule recommendation algorithm can explore the potential interests and association relationships of users, providing them with a certain degree of personalized recommendations. However, this algorithm has a relatively low degree of personalization and cannot meet the personalized recommendations of complex user needs., A hybrid recommendation algorithm is a synthesis of multiple recommendation algorithms in order to complement each other's strengths and improve recommendation accuracy and coverage. Although hybrid recommendation algorithms inherit the advantages of individual algorithms, their computation time and complexity also increase accordingly.

5.2 System design

As shown in Fig. 5, the database stores personal information and learning records of students, as well as relevant data on music knowledge and learning resources. The strategy layer is the decision-making and data analysis part of the system. It utilizes data in the storage layer to support personalized recommendation services of the system. In the strategy layer, the system conducts data mining and intelligent analysis based on the learning needs and interests of students by analyzing the learning process and learning outcomes. This analysis can be used for personalized services such as pushing learning resources, intelligent guidance, learning effectiveness statistics, and ability assessment.

5.3 Personalized recommendation module

In order to better understand and meet the needs of users, personalized recommendation systems use mining and organizing search records, as well as analyzing user behavior on the website. Through these analyses, the system obtains user preference information and selects data or products that meet user needs and possible preferences from a large amount of data, and then recommends them to users. The architecture of a personalized recommendation system is shown in Fig. 6.

The access and response between users and servers play an important role in recommendation functionality. This process is mainly used to organize and collect information related to users. The recommendation function can collect user information in two ways: explicit and implicit. Explicit approach is to obtain user feedback through explicit behaviors such as purchasing courses, listening to music, watching movies, reading books, etc. This method can obtain clear user preference information. Implicit approach is to obtain implicit data information of users by analyzing their behavior patterns and historical records, in order to predict their interests and needs. The recommendation function needs to dynamically follow the changes of users to obtain accurate information. The combination of explicit and implicit methods is the best way to achieve highly personalized recommendations. The recommendation results are returned to the user through the server and displayed in the recommendation interface.

Data mining and processing involve generating and training recommendation models, establishing multiple valuable recommendation models by mining user implicit information to improve personalized recommendation systems. Personalized recommendation includes personalized recommendation technology and related algorithms. By integrating these technologies and algorithms, personalized recommendation engines can provide accurate recommendation results and improve the efficiency of recommendations.

5.4 Analysis of teaching effectiveness

As shown in Table 3, the music learning ability of the experimental group learners and the control group learners showed an upward trend in all four tests. Both groups of learners showed an increase in their abilities during the first and second tests, but there was no significant difference. But in the third and fourth tests, the music learning ability of the experimental group learners began to be significantly higher than that of the control group learners, especially in the fourth test. The findings indicate that personalized resource recommendation based on learning analysis has a positive impact on improving the teaching ability of learners. After receiving personalized resource recommendations, the experimental group of learners further improved their music learning ability, ultimately reaching a higher level than the control group. This emphasizes the potential of personalized learning methods in improving learners' teaching abilities.

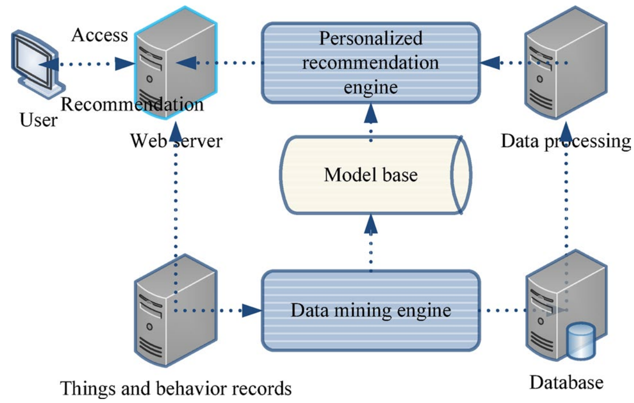


Fig. 6 Personalized recommendation algorithm set model

Table 3 Posttest results of teaching ability of experimental learners

Number of tests	Group	Divide equally	Standard deviation
1	Experimental group	70.06	12.92
	Control group	70.36	12.52
2	Experimental group	73.21	13.65
	Control group	72.64	13.90
3	Experimental group	73.82	7.33
	Control group	78.43	5.44
4	Experimental group	77.13	6.44
	Control group	82.10	4.76

6 Conclusion

Music education has always been an important discipline for cultivating students' musical abilities. However, the traditional sight singing teaching model has some limitations and is difficult to meet the needs of personalized teaching and self-directed learning. In order to improve this situation, the music teaching field has begun to actively explore teaching reforms under different conditions and levels. Multimedia technology is widely used in music teaching as teaching aids or auxiliary teaching systems. However, relying solely on multimedia technology cannot proactively provide personalized teaching services for students and teachers. In order to achieve personalized teaching of music sight singing, it is necessary to research a more intelligent and autonomous teaching system. Based on this, this article proposes a music information teaching and personalized recommendation system based on wearable sensing and soft computing algorithms. The design goal of this system is to provide assistance to researchers in the field of music education. Wearable sensing technology can capture the physiological and psychological states of students during the learning process, and apply this information to personalized teaching recommendations. Meanwhile, the application of soft computing algorithms can analyze and explain the learning characteristics of students, providing them with more accurate teaching suggestions and guidance. The advantage of this system is that it can provide personalized teaching content and recommendations based on the characteristics and needs of students. The music information teaching and personalized recommendation system based on wearable sensing and soft computing algorithms provides a new teaching method for researchers in the field of music education. I hope that the research and application of this system can contribute to the development of music education and promote the personalized and intelligent process of music teaching.

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Declarations

Conflict of interest The author reports no conflicts of interest.

Ethical approval This article does not contain any studies with human participants performed by any of the authors.

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